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ABSTRACT

The paleomagnetism of sedimentary rocks from the Huronian Gowganda Formation was studied for 16 sites near Elliot Lake, Ontario, in the Southern Geologic province. The rocks retain a stable remanence acquired during diagenesis some 2.29 b.y. ago. The calculated paleolatitude of 62° favors a polar glacial till mode of deposition for the thick tillite members, rather than a subpolar periglacial dropstone mode. The Gowganda pole at lat 65°N, long 123°W (d_p = 10.6°; d_m = 11.7°) indicates that either the ice cap was much larger than now thought or the Bear and Slave provinces were separated from the Churchill, Superior, and Southern provinces in Paleoaphebian time. The pole also requires a radical change in previously published Paleoaphebian polar wander paths.

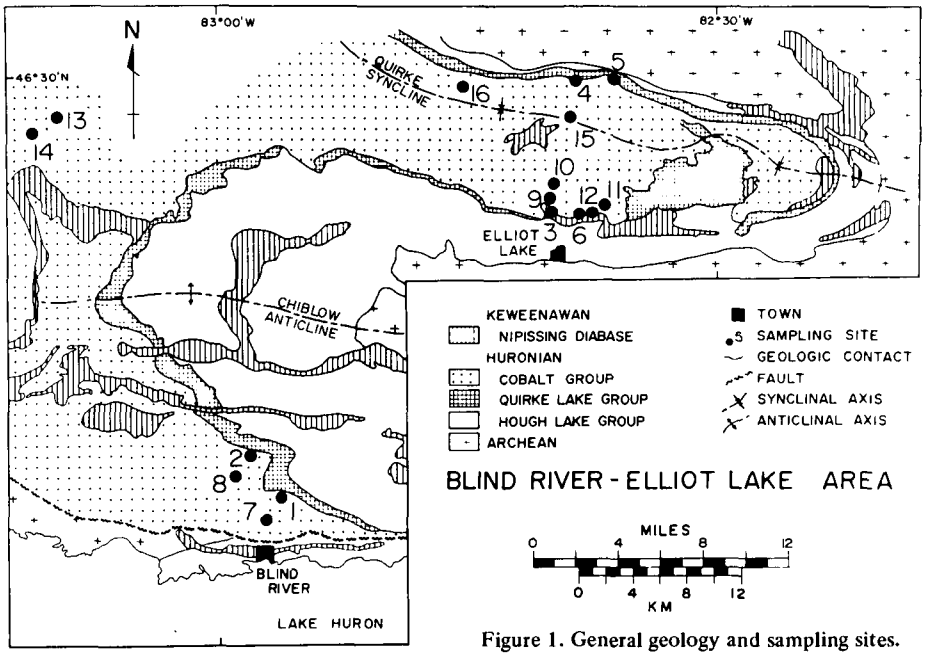


Figure 1. General geology and sampling sites.

Huronian glaciation and polar wander from the Gowganda Formation, Ontario

D.T.A. Symons

Department of Geology, University of Windsor, Windsor, Ontario N9B 3P4

INTRODUCTION

The origin of the till-like conglomerate of the Huronian Gowganda Formation is controversial (Roscoe, 1969). The argument centers on whether the conglomerate was deposited directly from the ice as a normal till in a glacial polar climate or whether the boulders were ice rafted and dropped into sediments deposited around an ice sheet in a periglacial subpolar climate. Paleomagnetic analysis provides a direct geophysical method to locate the pole position at the time of deposition and thus helps distinguish the hypotheses. Accordingly, cores from the Gowganda sedimentary rocks were collected at 16 sites near Elliot Lake, Ontario (Fig. 1), where the rocks have been altered by compaction and gently deformed.

GEOLOGY

The Gowganda Formation of the Huronian Supergroup crops out extensively in the Penokean fold belt of the Southern province to the west of the town of Elliot Lake and to the northeast in the Cobalt plate of the adjacent Superior province (Stockwell, 1969).

The Huronian Supergroup unconformably overlies an Archean basement composed predominantly of granitic gneiss with phases of metamorphosed alkalic igneous rocks and belts of metastrata. These rocks give radiometric ages of 2.45 to 2.7 × 10^9 yr (Roscoe, 1973).

The Huronian Supergroup is divided from bottom to top into the Elliot Lake, Hough Lake, Quirke Lake, and Cobalt Groups (Robertson and others, 1969). The lower part of the Elliot Lake Group contains alkalic volcanic rocks with silicic members such as the Thessalon Formation and feldspathic quartzites with uraniferous pyritic quartz-pebble conglomerate. The upper part contains argillaceous quartzite with siltstone and argillite. The remaining three groups are cyclic with basal conglomeratic graywacke or tillite units, overlain first by finer grained argillaceous units and in turn by cross-bedded arenaceous units. These three cycles are thought to reflect the advance, retreat, and isostatic adjustment stages of three glacial intervals (Roscoe, 1969; Frarey and Roscoe, 1970).

The Gowganda Formation is the lowermost formation or "advance" stage of the Cobalt Group. This formation is 1,000 to 1,500 m thick. It contains thick units of massive conglomeratic graywacke or tillite. The Archean tillite clasts are angular to rounded and range up to large boulder size. Some appear to be glacially striated (Roscoe, 1973). The presence of a striated basement under the Gowganda Formation in the Cobalt plate favors the "polar till sheet" genetic hypothesis. The Gowganda

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TABLE 1. REMANENCE DIRECTIONS BY SITES

Site*	Bedding		Cores+	Mean Remanence		
	Strike	Dip		Decl.	Incl.	α_{95}
1L	107°	23°S	4,4	337°	65°	5°
2L	100°	12°S	4,2	340°	82°	7°
3L	290°	8°N	4,2	246°	82°	14°
4L	85°	35°S	4,4	323°	77°	5°
5L	110°	35°S	5,3	336°	82°	15°
6L	289°	18°N	5,2	222°	79°	4°
7M	125°	25°S	4,4	334°	68°	12°
8M	90°	12°S	4,3	334°	69°	11°
9M	280°	11°N	5,5	322°	59°	5°
10M	290°	5°N	5,5	321°	76°	10°
11M	270°	7°N	5,5	320°	50°	5°
12M	270°	12°N	5,4	257°	74°	11°
13M	225°	5°N	3,3	347°	66°	5°
14U	225°	5°N	5,5	352°	59°	14°
15U	0°	0°	4,3	327°	87°	14°
16U	90°	30°S	3,3v	16°	82°	11°

* Gowganda Fm. - lower (L), middle (M), upper (U).
\+ Cores - collected, accepted.
v Polarity of one core at antiparallel position.

TABLE 2. REMANENCE DIRECTIONS BY GROUPS

GROUP	SITES	MEAN REMANENCE		
		Decl.	Incl.	α_{95}
LOWER	6	314°	81°	9°
MIDDLE	6	319°	67°	11°
UPPER	4	352°	74°	15°
ALL	16	326°	75°	6°
ALL COR*	16	302°	77°	10°

* Tectonic corrections made by rotating site mean directions through their bedding dip angle about their strike direction.

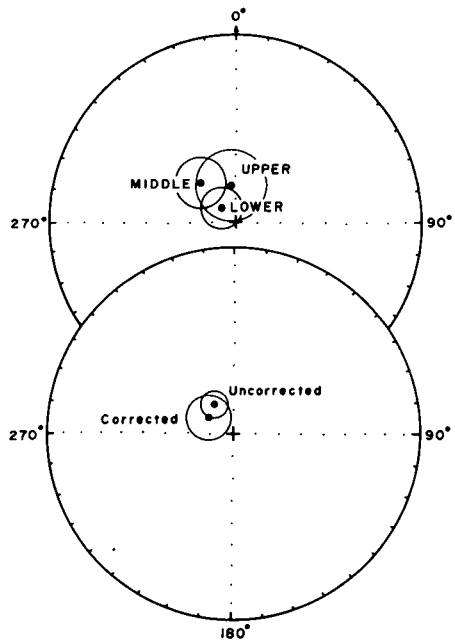


Figure 2. Equal-area stereographic projection nets giving the mean remanence directions surrounded by their cones of 95 percent confidence for the Gowganda Formation: (upper net) divided into lower, middle, and upper stratigraphic levels uncorrected for tectonic tilting, and (lower net) combining all sites both before and after tectonic correction.

Formation also has thick units of fine-grained siltstone, graywacke, arkose, and petromictic conglomerate occurring preferentially in the upper half of the formation. These units commonly contain laminated fluvial layers with occasional dropstones, which favors the "subpolar ice-rafted" hypothesis. The Gowganda sedimentary rocks are drab or very slightly reddish in color, indicating a reducing to neutral environment of deposition (Roscoe, 1969). Unmetamorphosed, flat-lying Gowganda sedimentary rocks from the Cobalt plate give a concordant Paleoaphebian Rb-Sr isochron age of deposition of 2.29 ± 0.09 × 10^9 yr (Fairbairn and others, 1969).

The Gowganda Formation near Elliot Lake was slightly deformed and altered during the Penokean orogeny. Dips rarely exceed 15° except at the margins of the Chiblow anticline and Quirke Lake syncline, where they reach 40° (Fig. 1). Alteration is shown mainly by diagenetic compaction and intergranular cementation. The Nipissing diabase was intruded during the Penokean orogeny (Church, 1966). The diabase gives a reliable Rb-Sr whole-rock isochron of 2.16 ± 0.08 × 10^9 yr, which dates the orogeny and sets a lower age limit for Huronian sedimentation (Van Schmus, 1965).

PROCEDURE

The 16 sampling sites come from the lower, middle, and upper stratigraphic levels of the Gowganda Formation. Care was taken at each site to distribute the samples vertically in different horizons as well as horizontally. The sampling was mostly confined to the Quirke Lake syncline and the Chiblow anticline to the north of Elliot Lake and Blind River, respectively, because the geology of both areas had been thoroughly mapped (Robertson, 1961, 1963, 1964, 1968). The sampling was also biased toward the flanks of the folds to provide a fold test (Graham, 1949) to distinguish between a prefolding or postfolding remanent magnetization (RM). Bedding attitudes were measured in the field or taken from the geologic maps.

At each site, three to five cores were drilled and oriented by sun compass. The natural remanent magnetization (NRM) was measured on a biastatic magnetometer (Larochelle and Christie, 1967) for two specimens from each core. One specimen per site with an average NRM direction and intensity was demagnetized in an alternating field (AF) (Larochelle and Black, 1965) in eight steps up to 800 Oe. Analysis of the RM directions on step demagnetization indicated that 200-Oe AF cleaning would succeessfully isolate the characteristic RM, and the 200-Oe intensity was used to clean the remaining specimens. For most specimens the RM intensities were between 5×10^{-6} and 5×10^{-7} emu/cm^3. The specimen RM directions were averaged to get the core RM directions. In 12 cores the two specimen directions diverged by more than 25°, and so these core RM directions were rejected. The core RM directions were then averaged to get the site mean RM directions (Table 1) with their cone of 95 percent confidence (a_{95}) (Fisher, 1953).

GOWGANDA REMANENCE

When the site mean RM directions are grouped by stratigraphic level, the level mean directions are not statistically significantly different (Table 2, Fig. 2). Thus all of the site means can be validly grouped into one population to calculate the "uncorrected" formation mean direction of 326°, 76° (a_{95} = 6°).

The fold test of Graham (1949) is inconclusive. The site mean RM directions were unfolded or corrected to bring the bedding planes to the horizontal by rotating about the strike direction through the dip angle. The resulting "corrected" mean direction is not statistically different from the "uncorrected" direction (Table 2, Fig. 2). Similarly, the slight increase in scatter of a_{95} values upon correction is not statistically significant. The simplest interpretation of the test results is that the primary depositional RM was mostly altered during diagenesis and folding to give the slight increase in scatter on unfolding, but this alteration happened shortly after deposition before the Earth's magnetic pole had time to change direction significantly.

Several other factors support the concept of a Paleoaphebian diagenetic

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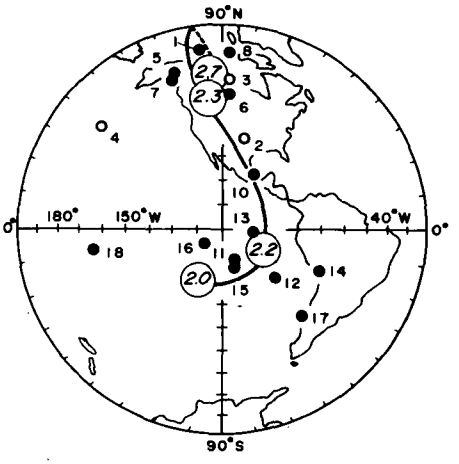


Figure 3. Polar wander path for Paleoaphebian rocks of North America. The poles used are (1) Archean gneiss (2.7 b.y.), Symons and Garber (1974); (2) and (3) Matachewan southwestern and northeastern dikes (2.69 b.y.), Fahrig and others (1965); (4) Dogrib dikes (2.69 b.y.), McGlynn and Irving (1974); (5) Thessalon volcanic rocks (2.35 b.y.), O'Leary (1974); (6), (7), (8), and (9) Gowganda Formation–lower, middle, upper, and overall (2.29 b.y.), this study; (10) Firstbrook Formation (2.29 b.y.), Symons 1967a); (11), (12), (13), (14), and (15) Nipissing diabase (2.16 b.y.), Symons (1967a, 1970, 1971), Patel and Palmer (1974), Symons and Londry (1975); (16) Indian Harbour dikes (2.08 b.y.), Murthy and Deutsch (1972); (17) and (18) Negaunee iron formation and ore (2.0 b.y.), Symons (1967b).

remanence. First, the sites include a variety of gray and reddish tillite, graywacke, siltstone, and arkose ranging through the stratigraphic section over a large geographic area, and all sites give similar directions. Second, demagnetization shifts the remanence directions away from the present Earth's magnetic field direction, leading to the field direction's exclusion from the cone of 95 percent confidence. Third, site 16 contains a core from one horizon with a reversed or antiparallel RM direction that is highly stable. Fourth, the pole position for the Gowganda is in good agreement with other Huronian poles, including the Thessalon volcanic rocks to the west (O'Leary, 1974; Fig. 3) and several Huronian sedimentary formations near Elliot Lake (D.T.A. Symons, unpub. data). The Firstbrook Formation pole (Symons, 1967a) is an exception. This formation crops out in the Cobalt plate and is equivalent to the upper Gowganda (Thompson, 1957). Its divergence may reflect limited paleomagnetic sampling both in area and in numbers. Also it has a reversed RM, so that it is not a strict time equivalent to the upper Gowganda sampled in this study. Fifth, the Gowganda pole is distinctly different from the younger Nipissing diabase poles from the Elliot Lake area (Symons, 1971; Patel and Palmer, 1974) and from concordant Nipissing poles from the Cobalt plate (Symons, 1967a, 1970; Symons and Londry, 1975). It is highly improbable that the Nipissing diabase would retain a primary RM while the enclosing Huronian strata would acquire a post-Nipissing metamorphic RM. Also, the distinct difference in Huronian and Nipissing poles suggests a marked time interval between diagenesis of the Gowganda sediments and subsequent intrusion of the Nipissing.

In summary the evidence suggests that the Gowganda sedimentary rocks acquired their RM some 2.29×10^9 yr ago. The pole position and paleolatitude of the Gowganda sedimentary rocks may then be used to discuss Paleoaphebian polar wander and glaciation.

PALEOAPHEBIAN POLAR WANDER

The Paleoaphebian polar wander paths proposed to date (Spall, 1971; Symons, 1971; Irving and Park, 1972) track southward from Nipissing pole positions west of Peru to Matachewan pole positions southwest of Cape Horn. The available pole positions for North American rock units in the 2.0- to 2.7-b.y. time interval are shown in Figure 3. The poles have been averaged into middle Aphebian (2.0 to 2.1 b.y., poles 16 to 18), Nipissing (2.16 b.y., poles 11 to 15), Huronian Paleoaphebian (2.2 to 2.4 b.y., poles 5 to 8, 10), and late Archean (2.4 to 2.7 b.y., poles 1 to 4) groups. The shortest path linking these four group means tracks northward from the Equator off Peru over the present Arctic into Russia, and this is considered the best estimate for the Paleoaphebian polar wander path. This path requires the pole to wander through approximately 120° of arc between Nipissing and late Archean times. The southward track, if the Paleoaphebian and late Archean poles are plotted in their antiparallel position, would require approximately 240° of arc, with the path moving from off Peru to the south Indian Ocean and back to southwest of Cape Horn.

Some comments on the reliability of these four group mean positions should be made. The middle Aphebian poles are poorly grouped, and both Negaunee poles are based on relatively few samples. Nevertheless the cluster fits as the logical continuation of polar paths developed for the 1.4-b.y. to 2.0-b.y. time interval (Spall, 1971; Irving and Park, 1972; Evans and

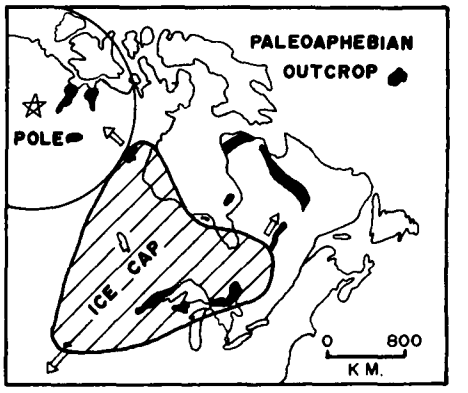


Figure 4. Position of the paleomagnetic pole for the Gowganda Formation relative to the Paleoaphebian ice cap of Young (1973).

Bingham, 1973; McGlynn and others, 1974). The Nipissing group mean is reliable because it is based on many sites over a large geographic area in the Superior and Southern provinces. The Huronian group is well clustered except for the Firstbrook Formation pole. The Animikie Gunflint iron formation pole at lat 28°N, long 94°W (Symons, 1966) should be included in this group on the basis of its location and stratigraphic correlation with upper Huronian formations by Young (1966). However, the Gunflint gives K-Ar dates of 1.90 ± 0.20 b.y. B.P. (Hurley and others, 1962) and Rb-Sr dates of 1.635 ± 0.024 b.y. B.P. (Faure and Kovach, 1969), which may be primary or may be reset by metamorphism (Misra and Faure, 1970). The late Archean group mean is somewhat doubtful. The Archean gneiss pole (pole 1) represents very few specimens. The Matachewan dikes may well be closer to 2.5 b.y. in age (Fahrig and others, 1965, and 1974, personal commun.), rather than 2.69 b.y. from Rb-Sr dating as given by Gates and Hurley (1973). Finally, the Dogrib dikes come from the Slave province. McGlynn and Irving (1974) suggested that the divergence between the Dogrib pole and Matachewan poles may reflect postemplacement rotation or translation between the Slave province and the rest of Laurentia. If so, then this pole should not be included with the other three poles, which come from the Superior province. It would, however, add weight to the argument given below that the Slave and Bear provinces were not attached to the Churchill-Superior-Southern "plate" during the Huronian glaciation.

PALEOAPHEBIAN GLACIATION

Using Holocene and Permian-Carboniferous glaciation limits as a guide (Flint, 1971; Hamilton and Krinsley, 1967;

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Smith and Hallam, 1970), a paleolatitude for the Gowganda tillite greater than 60° would strongly favor the polar glacial till hypothesis for their origin. Conversely, a paleolatitude of less than 40° would support the subpolar periglacial ice-rafted hypothesis. The determined value of 62° for the Gowganda indicates a polar paleoenvironment and thus favors the glacial till hypothesis.

The Gowganda pole is located at lat 65°N, long 123°W on the south shore of Great Bear Lake, where Precambrian basement of the Bear province is overlain by a thin veneer of flat-lying Phanerozoic sedimentary rocks. This pole is not within the limits of the Paleoaphebian ice cap that Young (1973) has determined from the stratigraphic record (Fig. 4). If we accept these limits, then the ice cap was eccentric with respect to the rotational pole—assuming it is axial with the magnetic pole. This is analogous to the Holocene ice caps of North America and Eurasia (Flint, 1971) and suggests by analogy that the Paleoaphebian pole was centered in an ocean. If true, then the paleoplate including the Churchill, Superior, and Southern provinces was separated from the Bear and Slave provinces during glaciation. The northwest transport directions indicated by paleocurrent analysis of the Hurwitz Group (Bell, 1969) on the northwest side of the ice cap and distribution of Paleoaphebian tillite favors this possibility. Alternatively, if we do not accept Young's limits, then the ice cap would have been much larger and centered in the Bear province. If true, then the Slave and Southern provinces were part of the same paleoplate. The presence of nonglacial Paleoaphebian sedimentary rocks in the Bear and Slave provinces that apparently correlate with those from the other provinces favors this alternative.

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